Thermal Expansion

Aim

To determine the coefficient of thermal expansion of various materials.

Analysis

All materials expand when heated and contract when cooled. For solids close to room temperature, the dependence is linear and can be described by the following equation.

$$\Delta L = L_0 \alpha (T - T_0)$$

Rearranging terms we get

$$\frac{1}{\alpha} = L_0 \cdot \frac{dT}{dL}$$

If f is a product of x_is then the uncertainty in f is given by

$$\left(\frac{\sigma_f}{f}\right)^2 = \sum_{i=1}^{n} \left(\frac{\sigma_{x_i}}{x_i}\right)^2$$

The error in α is given by

$$\sigma_\alpha = \sqrt{\left(\frac{\sigma_{L_0}}{L_0}\right)^2 + \left(\frac{\sigma_m^2}{m}\right)^2}$$

We calculate this for different materials.

Safety precautions : The induction stove must not go beyond 800 watts. The pipes carring steam must not be bent. Use gloves to touch hot equipment.

Brass

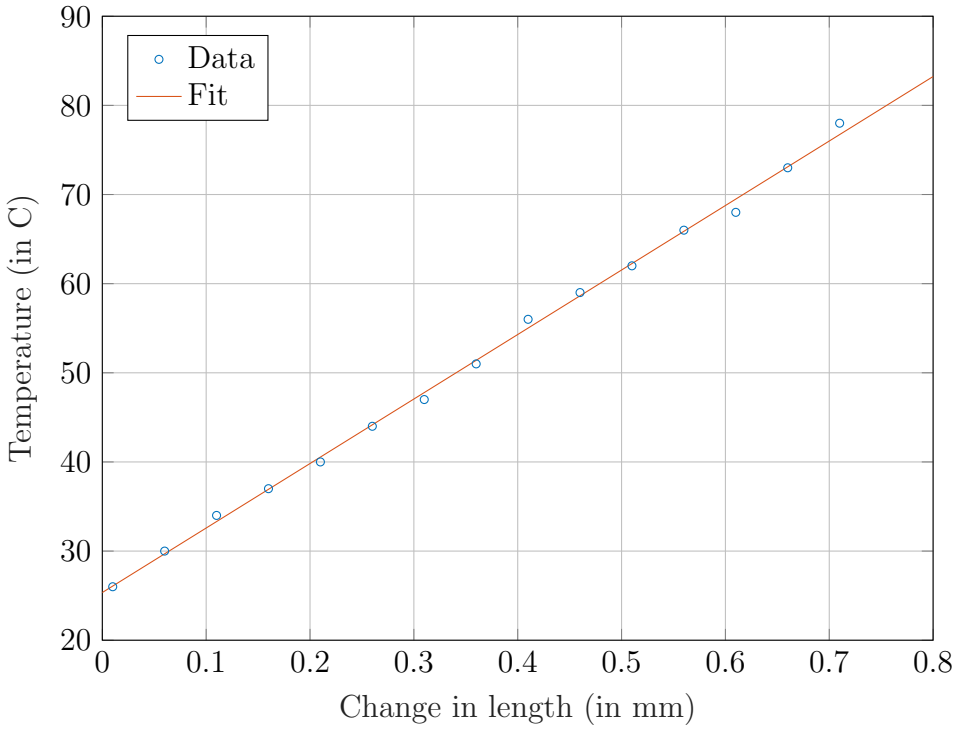


Figure 1: The slope is 71.36 ± 0.85

$$\alpha = \frac{1}{582 \times 71.36} = 2.408 \times 10^{-5}$$

The uncertainties are

$$\sigma_{L_0} = 1$$
$$\sigma_{slope} = 0.85$$

Thus we get

$$\sigma_\alpha = 1.203\% = 0.029 \times 10^{-5}$$

The coefficient of thermal expansion in brass is $2.408 \pm 0.03 \times 10^{-5}$

Copper

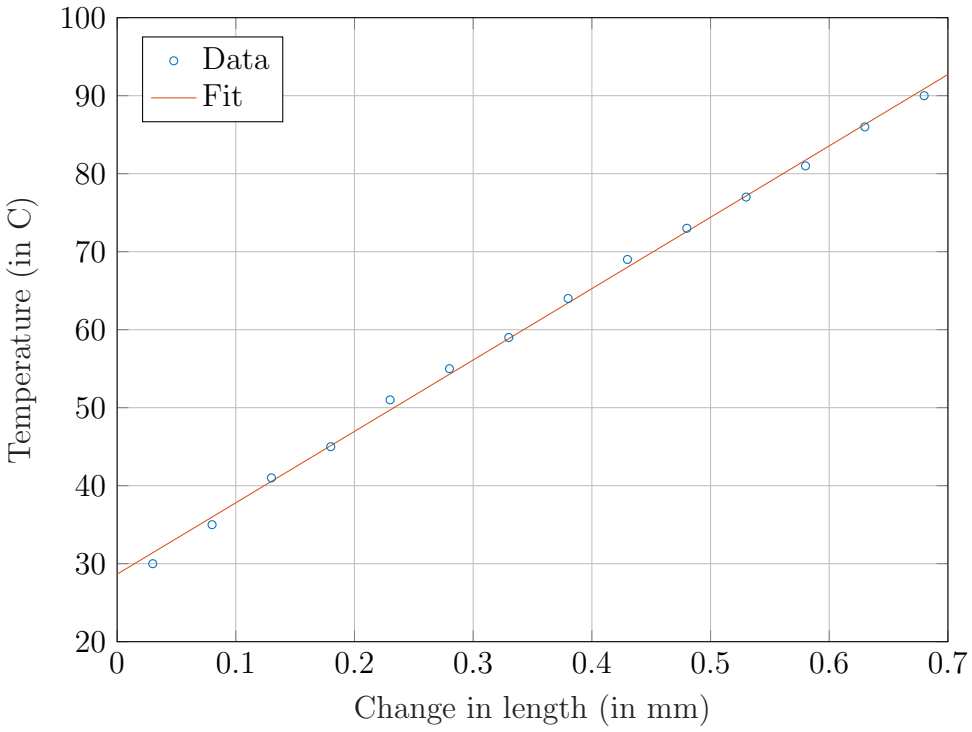


Figure 2: The slope is 91.51 ± 1.10

$$\alpha = \frac{1}{582 \times 91.51} = 1.878 \times 10^{-5}$$

The uncertainties are

$$\sigma_{L_0} = 1$$
$$\sigma_{slope} = 1.10$$

Thus we get

$$\sigma_\alpha = 1.215\% = 0.023 \times 10^{-5}$$

The coefficient of thermal expansion in copper is $1.878 \pm 0.023 \times 10^{-5}$

Aluminium

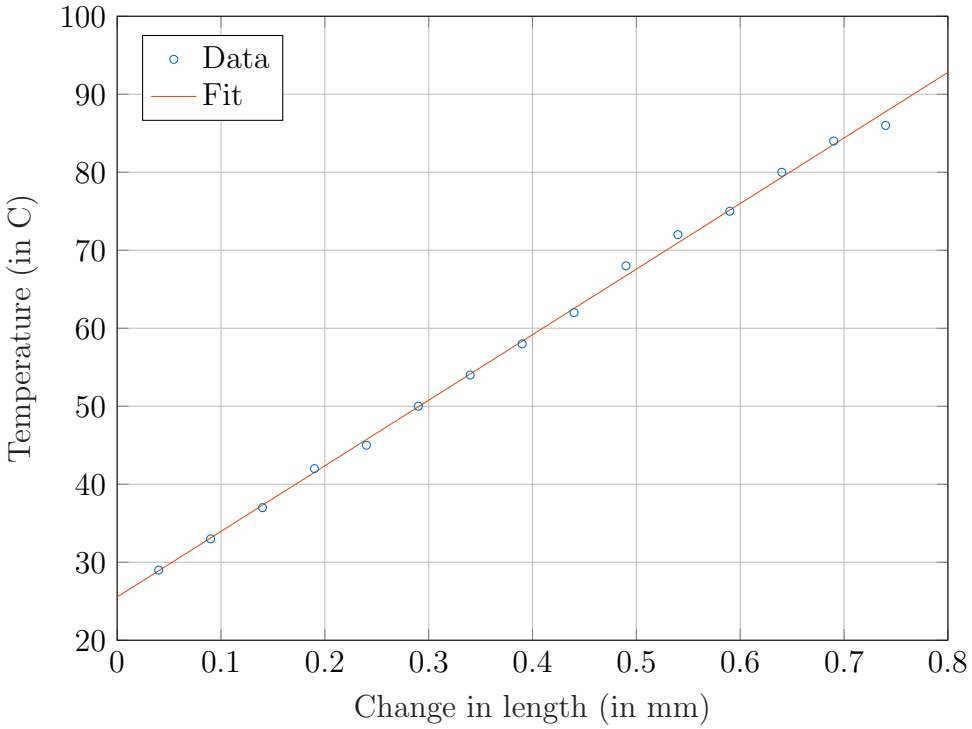


Figure 3: The slope is 84.07 ± 0.92

$$\alpha = \frac{1}{582 \times 84.07} = 2.044 \times 10^{-5}$$

The uncertainties are

$$\sigma_{L_0} = 1$$
$$\sigma_{slope} = 0.92$$

Thus we get

$$\sigma_\alpha = 1.110\% = 0.023 \times 10^{-5}$$

The coefficient of thermal expansion in aluminium is $2.044 \pm 0.03 \times 10^{-5}$

Result

The values of coefficient of thermal expansion for different materials is listed below.

$$\begin{aligned} \text{Brass} &: 2.408 \pm 0.029 \times 10^{-5} \\ \text{Copper} &: 1.878 \pm 0.023 \times 10^{-5} \\ \text{Aluminium} &: 1.110 \pm 0.023 \times 10^{-5} \end{aligned}$$